Microsoft Active-Directory home lab

1. Home lab setup

This lab consists of 2 laptops connected through home wifi router (both laptops part of same LAN)

Figure 1: a photo of the home lab setup.

Figure 2: a rough diagram of how —---

1.1 Network connectivity

PC 2 and PC 3 are connected through home wifi router

- IP address of PC2: 192.168.8.193
- IP address of PC3: 192.168.8.148

1.2 How logs are relayed from windows 10 VM(PC1) to splunk indexer running on PC3

1. PC1 sends logs to PC2 through isolab
2. PC2 relays logs to PC3 through home wifi router

alias	What it represents	Operating system	Purpose of this PC	Installed software
PC 1	Windows 10 VM (guest machine)	Windows 10	This is the victim PC that generates malicious telemetry	1. Sysmon 2. Splunk universal forwarder
PC 2	Laptop #1	Lubuntu OS	Host windows 10 VM & forwards its logs to Laptop #2	Splunk universal forwarder
PC 3	Laptop #2	Windows 11	Runs splunk indexer, indexes logs and hosts splunk dashboard	Splunk enterprise indexer

2. Goals

The goal of this lab is not to learn AD to a level I'd become an Active Directory administrator. But to be familiar with AD to a point i can confidently understand the topics that I would come across when I'm a SOC analyst. I got the topics from CHATGPT (and the topics that were trivial or unrelated were later removed by me). What tasks to perform in this lab were decided by me.

1. Understand connection between Windows server and Active directory.
2. Understand why AD requires DNS server role?
3. Learn the AD topics SOC analysts come across :
    - Users and groups
    - Domain Controllers
    - Domain vs workstation
    - OUs
    - Group Policy
    - Authentication
    - Kerberos
    - NTLM
    - LDAP
    - DNS and AD
    - Security groups
    - Privileged accounts
    - Service accounts
    - Domain Admins
    - Computer accounts
    - Windows Event Logs
    - Security Event IDs
    - PowerShell
    - Windows authentication failures
    - Account creation/deletion
    - Group membership changes

2.1 Goal1 - Understand connection between Windows server and server roles

When we install windows server we get 3 things

1. The operating system
2. Server roles (all are disabled by default)
3. Server features

- Active Directory Domain Services (AD DS)
- DNS Server
- DHCP Server
- File and Storage Services

- Hyper-V
- Web Server (IIS)
- Remote Desktop Services
- Active Directory Certificate Services (AD CS)
- Active Directory Federation Services (AD FS)

2.2 Goal2 - Understand AD is and why it requires DNS

AD uses DNS to:

- Locate Domain Controllers
- Locate services such as LDAP and Kerberos
- Allow domain-joined computers to find AD services
- Support authentication and domain operations

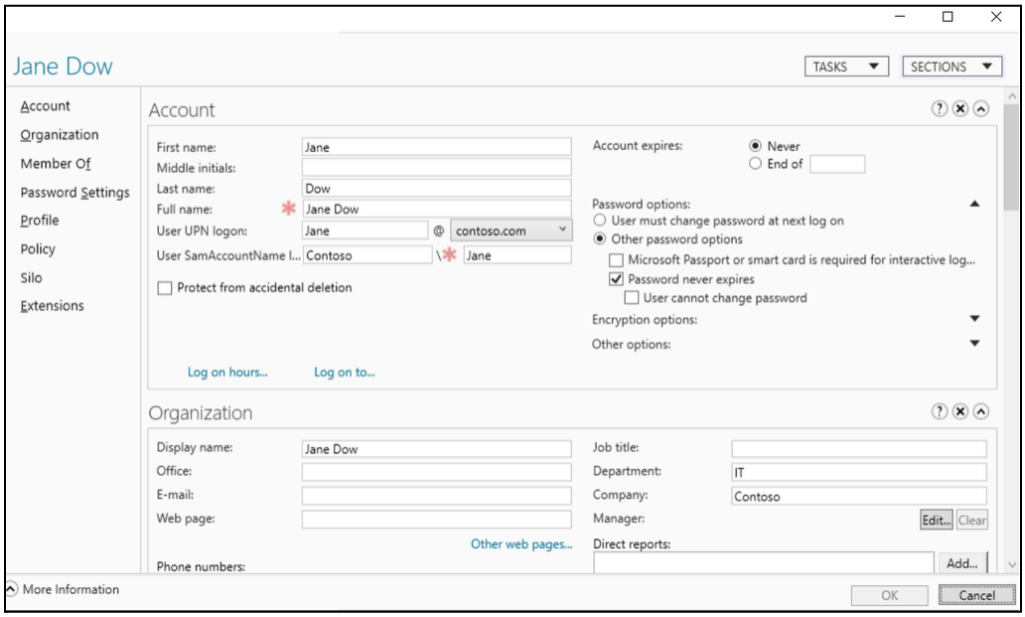
2.3 Goal3 - Learn the AD topics SOC analysts comes across

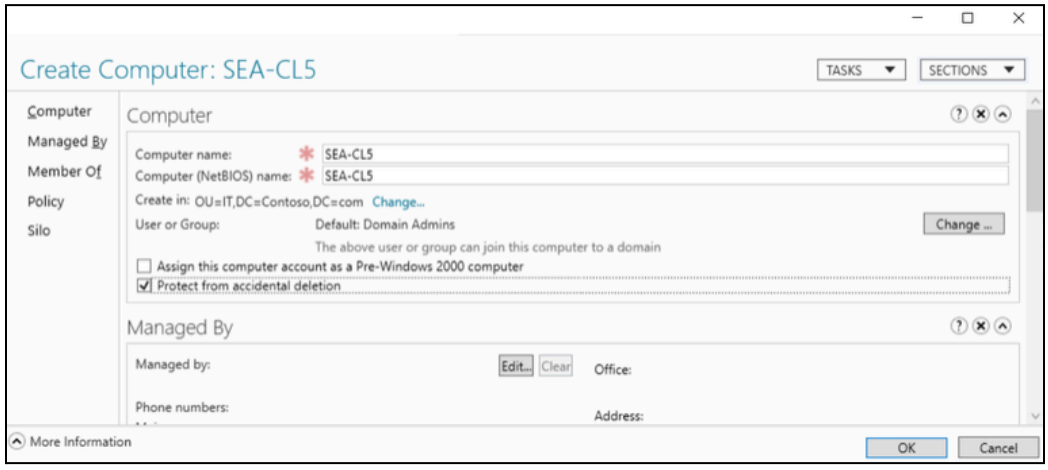
1. What is AD DS?
    - The AD DS database is the central store of all the domain objects (user accounts, computer accounts, and groups). AD DS provides a searchable, hierarchical directory (which means information is organized like folders).
2. What are AD DS components
    - Logical components - Domain, Organizational Units, Users, Groups, Forests, Schema, Containers, Partition
    - Physical components (things that represent real world objects) - Data store, Domain controller, Site, Subnet.

Eventhough user object represents real users, the object itself is just logical information stored in Active directory.

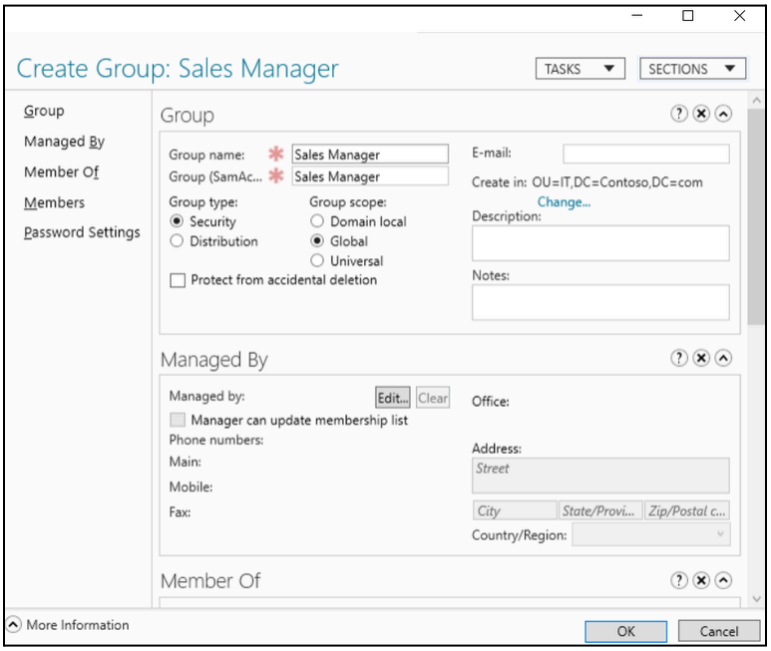
3. What is the relationship between Forest,Domain, AD objects and DNS
    - Domain is like a administrative/Security area which contains AD objects such as Users, Groups, Computers, OUs.
    - The domain has a DNS name such as Michelin.com
    - This allows computers to search for that domain's domain controller
    - Inside each domain we have Organization units such as finance, IT, HR
    - Inside those OUs we have users, computers and groups (these are the low level objects)
    - Forest is the top-level logical structure of an Active Directory environment.It can contain one or multiple domains. For example Michelin.com might have different

services. In that case the organization need to manage user accounts, computers of all the different sectors. Therefore the org can make domains for each sector such as tyres.michelin.com and restaurants.michelin.com .

4. Account types in AD DS?
    - User accounts
        - When users need to authenticate themselves.
        - 

        - Auser account is an object that contains all the information that defines a user. A user account includes: The username, A user password and Group memberships. The username and password of a user account serve as the user's sign-in credentials
    - Managed ervice accounts (MSA), Group Managed Service Accounts (gMSA), Delegated Managed Service Account (dMSA).
        - When services need to authenticate themselves.
    - Computer accounts
        - When computers need to authenticate themselves
        - 

    - Group objects

- permissions to user accounts arent added directly, they are added to groups and users are then assigned to those groups)
- 

- Group type can either be security or distribution. Group scope can be either domain local, local, global or universal.

5. Why are service accounts needed.
    - Services running in OS might also need to authenticate themselves. Some organizations just make a domain for services and give user accounts to them. This works but password management becomes a mess. Better approach is using MSA or even specilaized MSA account such as gMSA and dMSA.

References

1. AD basics Mircosoft self learning module: https://learn.microsoft.com/en-us/training/modules/introduction-to-ad-ds/